Machine Learning - Undergraduate Level Assessment 7 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Statement 1| Overfitting is more likely when the set of training data is small. Statement 2| Overfitting is more likely when the hypothesis space is small.

   (a) True, True

   (b) False, False

   (c) True, False

   (d) False, True

2. Which one of the following is equal to P(A, B, C) given Boolean random variables A, B and C, and no independence or conditional independence assumptions between any of them?

   (a) P(A | B) * P(B | C) * P(C | A)

   (b) P(C | A, B) * P(A) * P(B)

   (c) P(A, B | C) * P(C)

   (d) P(A | B, C) * P(B | A, C) * P(C | A, B)

3. Which one of the following is the main reason for pruning a Decision Tree?

   (a) To save computing time during testing

   (b) To save space for storing the Decision Tree

   (c) To make the training set error smaller

   (d) To avoid overfitting the training set

4. MLE estimates are often undesirable because

   (a) they are biased

   (b) they have high variance

   (c) they are not consistent estimators

   (d) None of the above

5. Averaging the output of multiple decision trees helps _.

   (a) Increase bias

   (b) Decrease bias

   (c) Increase variance

   (d) Decrease variance

True / False (10 marks)

Answer True or False

6. True or False: In Yann LeCun's framework of machine learning, reinforcement learning is considered the cherry on top.
7. True or False: The dimensionality of the null space of the matrix A is 1.
8. True or False: Increasing the number of training examples always results in higher variance in the model's predictions.
9. True or False: Predicting the amount of rainfall in a region based on various cues is a supervised learning problem.
10. True or False: RELU activation functions are monotonic, while sigmoid functions are not monotonic.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find number of lists present in the given tuple.
12. Write a function to find the largest possible value of k such that k modulo x is y.

End of Paper